An Input-Exposure Control for Ontology Grounded Generation over Private Corpora

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The Loom / VisionFlow neurosymbolic stack

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Abstract

An organisation that grounds an AI assistant in its own curated knowledge gets its accuracy from the knowledge rather than the model: shown the right vetted facts, a small locally hosted model answers domain questions as well as the largest commercial systems. This paper builds the instrument that makes that claim provable rather than assumed.

When a language model answers from a private, curated corpus via graph-based retrieval, the reported "grounding uplift" obscures a measurement issue: if gold answers come from the same injected corpus, uplift mixes faithful *delivery* of exposed facts with *reasoning* over injected structure. We propose the *copy ceiling*, defined as the recall a verbatim copy of the shown context would achieve, and the signed *gain over copy*. This control is deterministic, judge-free and per-item, listed as a standing column beside generation recall, and paired with a placebo verified irrelevant by seed-disjointness; together they separate exposure coverage from beyond-exposure recovery prior to architecture selection. On a production ontology-serving node, unaided models score ≈ 0.26 in-domain and grounded ≈ 0.92, yet across ten models from five providers the gain over copy is uniformly negative (-0.067 to -0.022). A direct exposure/recovery decomposition of 11,360 gold items explains why: three unexposed gold items are recovered in total, context utilisation spans 0.900–0.955, and models differ only in how much exposed gold they drop. The uplift is delivery, measured rather than inferred. A paired study on the same node, holding the model constant and varying only the serving path, lifts judged quality by +0.27 pooled ($p = 0.004$) and +0.79 where curation is deepest; a four-arm control study is consistent with content-specific transfer (the served block lifts quality, +0.59; the verified-irrelevant placebo's point estimate is near zero, +0.04, though differential attrition leaves the contrast imprecise). Out of domain the gated node does not regress: the general-set delta (+0.05, [0.00, +0.13]) lies inside a pre-specified ± 0.25 equivalence margin. Grounding evaluations whose gold derives from the injected corpus should report a copy ceiling as standard.

1 Introduction

Enterprises need language models to answer from private, domain-specific corpora absent from pretraining. Retrieval-augmented generation bridges parametric memory and required knowledge [10, 23], particularly for long-tail and proprietary facts [29]. Structured, graph-based retrieval further sharpens this approach [3, 6, 14, 32]. Yet when gold answers derive from the injected corpus, headline "grounding uplift" conflates faithful *delivery* of exposed facts with *reasoning* over injected structure. Published knowledge-graph grounding reports lack a standing control that separates them.

We introduce the *copy ceiling*[1], a per-item, graph-derived input-exposure baseline, and the signed *gain over copy*. The copy ceiling is the recall that a no-op extractor of exactly the shown text would achieve; gain over copy is grounded recall minus this ceiling. This deterministic, judge-free control is paired with a verified seed-disjoint placebo. It generalises the discrete input-only baselines used to interpret reading comprehension [19] and natural-language inference [15, 36]. Grounding evaluations whose gold derives from the injected corpus should report it as standard [21, 30].

Without assistance, the evaluated models score ≈ 0.26 on genuinely private in-domain questions and ≈ 0.92 with curated context. Across ten models from five providers, gain over copy is uniformly negative (-0.067 to -0.022), while beyond-exposure recovery totals three items in 11,360 (§7.2). Models therefore differ almost entirely in how much surfaced gold they drop. High-fidelity delivery of an authoritative, human-vetted source is the product: answers inherit its trustworthiness, are attributable to a named generation, and amortise curation rather than repaying it per query.

Contributions.

1. **The protocol (C1, singular).** A formally defined deterministic per-item exposure reference, its signed gain over copy, an exposure/recovery decomposition separating dropped exposed facts from beyond-exposure recovery, and a placebo verified irrelevant by construction, with explicit assumptions (§5).
2. **The evidence (C2).** Ten models are uniformly negative on gain over copy; a four-arm negative-control design (true / shuffled / masked / irrelevant scaffolds) tests content specificity; and a production-node paired study isolates the ecosys-

[1] The term *copy ceiling* is used independently and in an unrelated sense by Liu [27] (a decoding-layer editing mechanism, where it denotes token reachability under a line-level copy primitive; that paper was withdrawn in 2026), which we flag only to head off a terminology collision.

tem feature by holding the model constant and varying only the serving path (§6; results in §7).

3. **The organising frame (C3).** A conjunctive multivariate bar combining in-domain delivery fidelity with out-of-domain non-regression, measured on separate instruments, with the copy ceiling as the in-domain axis (§2, §11).
4. **The artefacts (C4).** An open harness (paired bootstrap, Wilcoxon, matched-pairs rank-biserial, Holm), frozen question sets, all per-row data, and the served system itself.[2]

2 The Private-Corpus Setting

The measured design choices follow from deployment requirements, each determining a design outcome and architectural decision record (ADR); Table 1 shows where each constraint is enforced. The system is a served node[3] operating over a corpus produced by a separate, CI-gated pipeline[4] and accelerated by an in-process HNSW index[5].

2.1 The ecosystem in brief

This report examines one component of the working VisionFlow stack. Knowledge enters through the *knowledgeGraph* repository: 8,138 Logseq markdown pages compiled losslessly into a pure-TBox OWL 2 ontology comprising 8,146 classes and no individuals, then published as versioned generations. Content comes from human-directed authorship and, more recently, automated extraction. Output from OntoCast [13], an independently developed open-source ontology-extraction tool that is not our work, enters through a preview-first import that reports identity collisions, refuses overwrites and emits private review candidates rather than publishing machine output. OntoCast remains an upstream producer at a governed boundary; neither project vendors the other. A CI gate combines consistency tests and a native conflict detector with a Whelk OWL 2 EL reasoner [4], classifying each build and rejecting contradictions before publication. The generation includes its reasoned closure of 282,492 triples.

The measured *Loom* node mirrors and serves a generation through lexical retrieval over class titles, confidence-gated injection of per-IRI markdown blocks, read-only SPARQL over the closure and delegation to the local model through an OpenAI-compatible façade, currently Qwen3.8-27B on two workstation GPUs. *RuVector* provides corpus embeddings in Postgres and an in-process HNSW semantic fallback. The fallback is disabled by default because its measured recall of 0.816 is below the 0.87 design floor. Downstream consumers, including an email gateway and working agents, inherit grounding by holding the façade URL. This context identifies the subject of the deployed-system measurements in §4–§7. Figure 1 shows the full path.

1. **Private-knowledge need ⇒ curated, amortised corpus.** The model must answer over a corpus it cannot have memorised (raw ≈ 0.26 is consistent with absence from pretraining, not proof). A human-directed, provenance-tracked ontology incurs curation cost once and is reused per query. The copy ceiling expresses at evaluation time that the curation carries the answer (ADR-135 D2; ADR-136 D4).
2. **Trust ⇒ human auditability at single-entity granularity.** A human must be able to audit served knowledge end-to-end. The canonical unit is therefore one per-IRI markdown-with-ontology block, comprising curated prose headed by typed relations, and every retrieval port returns or resolves to an `Iri` addressing it (Invariant I-P1; ADR-136 D1).
3. **Auditability ⇒ accelerators strictly behind the markdown.** The lexical index, HNSW, oxigraph SPARQL and attestation ledger only find, rank or attest the markdown unit and resolve to its IRI. An acyclic crate ring enforces this boundary (ADR-136 D1–D3).
4. **Privacy / LAN-confinement ⇒ local delegation, no cloud on the hot path.** Content and answers remain on the local network; generation is delegated by URL to a LAN/local backend, and augmentation is fully in-process (ADR-135 D1; ADR-136 §3).
5. **Local-model churn ⇒ model swap without consumer change.** Local models change often (Gemma → Muse-Glimmer → Qwen3.8-27B). A stable OpenAI-compatible façade makes the backend one configuration line and records model identity in the result rather than the endpoint, allowing this study to hold grounding constant while varying the model (ADR-135 D1).
6. **Bounded latency ⇒ retrieval works with no model, fast.** The scaffold, SPARQL and search endpoints require no model call; the lexical inverted index resolves in **2.02 ms** at the median over 8,146 class titles (ADR-136 §3).
7. **Injection interference ⇒ confidence-gated, benchmark-first injection.** Context can displace correct parametric knowledge on off-ontology queries [28, 39, 41]. A single injection authority scales the budget to the retrieval score, skips below a threshold [29], and benchmark-gates features that could add noise (ADR-136 D3).
8. **Never-mixed provenance ⇒ attestation and boundary discipline.** Every grounded answer

[2]The Loom serving node: https://github.com/DreamLab-AI/loom.

[3]The served node https://github.com/DreamLab-AI/loom (code under the repository licence; corpus terms per the sibling knowledgeGraph repository).

[4]The canonical builder https://github.com/DreamLab-AI/knowledgeGraph (published at https://narrativegoldmine.com) runs `pytest pipeline/tests` and `pipeline.validate` before publishing each generation.

[5]RuVector: https://github.com/ruvnet/ruvector.

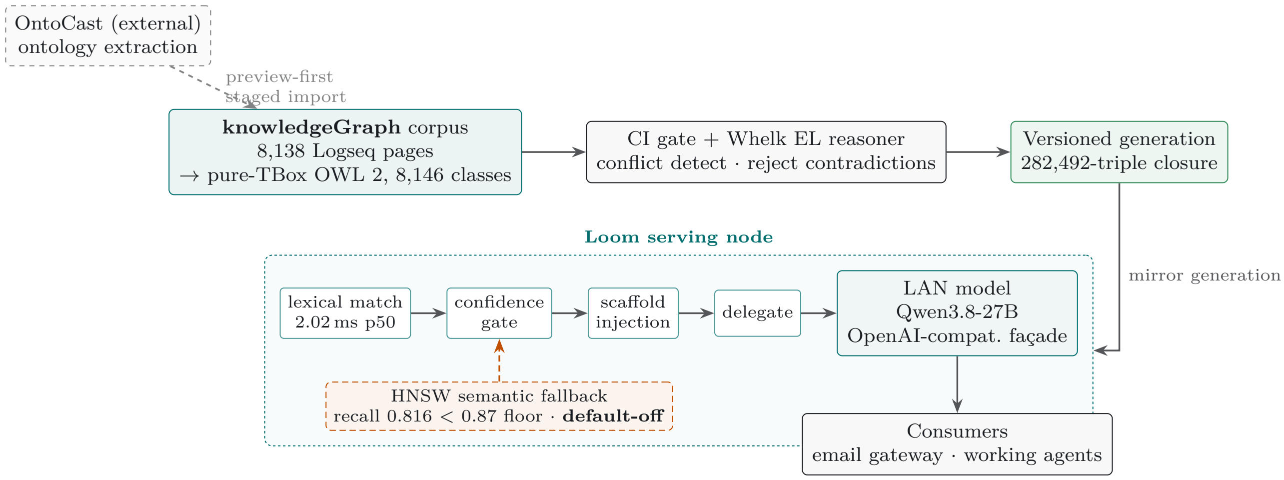

Figure 1: The data path around the measured node. Knowledge enters at the *knowledgeGraph* corpus (8,138 Logseq pages compiled losslessly into a pure-TBox OWL 2 ontology of 8,146 classes); output from OntoCast, an independent third-party extractor, enters only through a preview-first staged import that reports collisions and emits review candidates rather than publishing machine output directly. A CI gate and a Whelk OWL 2 EL reasoner classify each build and reject contradictions before publishing a versioned generation (a 282,492-triple reasoned closure). The *Loom* node mirrors a generation and serves it through a model-free pipeline (lexical match at 2.02 ms, confidence gate, scaffold injection) before delegating to whatever LAN model sits behind its OpenAI-compatible façade (Qwen3.8-27B here); consumers hold the façade URL and inherit grounding. The HNSW semantic fallback is drawn dashed in burnt orange because it ships default-off: its document-embedding recall (0.816) is below the 0.87 design floor, a red gate we report rather than hide (§2.1, §4).

traces to a named, versioned generation and carries a `loom` telemetry block and `corpusNature` honesty metadata. The node serves only the published ontology, never the working graph (ADR-135 D6/D7; ADR-136 D5).

These requirements jointly demand attributable recall where the corpus is authoritative and no regression on general questions the model already answers. We reconcile these axes in §11.

3 Related Work

Knowledge-graph and private-corpus grounding. The lineage runs from RAG [23] through graph-structured retrieval [6, 14] and zero-shot knowledge-graph triple prompting [3] to roadmaps and surveys [32, 35]. GraphRAG addresses "private and/or previously unseen" collections [6], while enterprise variants highlight amortised curation [40]. Published work typically injects *instance* triples or opaque encodings, including community summaries [6] and GNN soft-prompts over retrieved subgraphs [16]. We instead inject human-readable, schema-level content: class definitions, typed relations, and taxonomy. None of these studies reports an input-exposure control, so their uplift remains uncontrolled for content already present in context.

Baselines that reframe a result. Simple controls can overturn headline results: strong RAG baselines match elaborate pipelines under equal token budgets [21]; controlled comparisons reverse earlier "RAG wins" findings and add a confidence router [25]; and apparent argument comprehension can reflect spurious cues [31]. The confidence router of Li et al. [25] and selective retrieval [2, 18, 29] provide precedents for our injection gate. Those methods choose whether to retrieve; ours chooses whether to inject.

Input-only baselines and faithfulness metrics. The copy ceiling extends input-only baselines. Passage-only, question-only [19], and hypothesis-only baselines [15, 36] reveal shortcuts present in inputs. Unlike these discrete task classifiers, the copy ceiling is continuous and per-item. Faithfulness can be measured separately from correctness [1, 8, 43]; metric-first evaluations formalise assumptions [30] and test shortcut robustness [9]. FActScore's source precision and Adlakha's token-overlap K-Precision resemble our exposure count. ALCE rejects "copy the passage" as a shortcut; we use it as the reference point. Contamination audits [12, 38] assess input-visible score post hoc, whereas we define deliberate exposure as the reference a priori.

The nearest neighbours, and how we differ. The closest methodological precedent causally manipulates the gold answer to test its effect on retrieval-augmented gains [24]. We adopt only its paired-intervention *method*: the authors withdrew the work in 2026 after finding errors that compromised its main conclusions. Leakage-free benchmarks instead extract reasoning graphs from question–context pairs, then apply type-constrained entity replacement with consistency checks and leakage filters [26]. This occurs during corpus construction; the copy ceiling is a per-item measurement. RAGChecker uses LLM extraction and grading to decompose RAG quality into claim-level retrieval

Table 1: The private-corpus requirements arc: each business requirement, the design consequence it forces, and the point at which the consequence is enforced (not merely intended). ADR = the governing architectural decision record.

Requirement	Design consequence	Enforcement point
Private, unmemorised corpus	Human-curated ontology, curation amortised once	Sha-addressable generation; raw ≈ 0.26 (ADR-135 D2)
Human auditability	Per-IRI markdown-with-ontology as the served unit	Ports resolve to `Iri` → `CanonicalUnit` (I-P1)
Accelerators behind the unit	Indexes only find/rank/attest the markdown	Acyclic crate ring; adapter cannot mint the unit
LAN confinement	Model delegated by URL; in-process hot path	`DISTILL_BACKEND_URL`; Profile A network-free
Model churn	Stable façade; model identity in the result	OpenAI-compatible `/v1`; one config line
Bounded latency	No-model retrieval path	Lexical index 2.02 ms p50; scaffold needs no model
Injection interference	Confidence-gated selective injection	Single injection authority; skip below threshold
Never-mixed provenance	Per-answer attestation; ontology-only boundary	`loom` block + `corpusNature`; working graph never served

and generation scores [37]. Our context utilisation is a deterministic, surface-matched version of its context-utilisation metric; we claim no originality for the decomposition. MIRAGE measures benchmark-level adaptability by evaluating reader behaviour against oracle context across retriever–model pairings [34]. The RAGAS *library*'s later "context recall", absent from the original paper [8], is an LLM-judged retrieval-coverage measure. We have not found elsewhere our combination of a deterministic, judge-free, continuous per-item exposure ceiling for a schema-level ontology corpus and a verified seed-disjoint placebo. Across models, signed gain over copy provides an exposure-normalised recall scale where raw uplift does not distinguish models. Judged arms use a cross-family judge to limit self-preference [33, 42].

4 The Ontology Scaffold

The deployed serving node is a single static Rust binary comprising an eight-crate hexagonal workspace: a pure `loom-domain` core defining the port traits, five adapters, a `loom-scaffold` policy crate, and a thin `axum` façade. This structure enforces the audit invariant in §2: only the domain crate generates a `CanonicalUnit`, so an adapter cannot return its own row, triple, or vector as the served payload (Invariant I-P1). Golden fixtures pin retrieval, gating, and scaffold serialisation byte-for-byte to a reference implementation.

The canonical unit and the corpus. The unit of service and review is a markdown-with-ontology block scoped to one IRI. Each block contains curated research prose preceded by typed relations: `subClassOf`, `requires`, `enables`, `uses`, `relatedTo`, and `contrastsWith`. The corpus is a single sha-addressable *generation*; the evaluated generation comprises **8,146 concept classes** and **282,492 triples** in the Whelk-reasoned closure. Atomic mirroring ensures that the node serves a known, mixed-build-free version without embedding data in the binary. Unlike approaches that reduce knowledge to opaque community summaries or GNN-encoded subgraphs [6, 16], this system retains reviewable markdown as the durable asset; the accelerators only locate, rank, or attest to it [11].

The lexical matcher and confidence gate form the exclusive injection authority. A single policy scales the scaffold budget to the retrieval score and injects nothing below a defined threshold, preventing erratic outputs on off-ontology queries (§7.5). The graph store runs read-only, LIMIT-clamped SPARQL over the reasoned closure without a model call. An in-process HNSW fallback over 384-dimensional embeddings supports out-of-vocabulary queries but ships disabled: its document-embedding recall of 0.816 misses the 0.87 design floor. It will remain off until a query-shaped embedding clears a multivariate benchmark, rather than through threshold relaxation; §12 revisits this red gate. Generation is delegated through the model-swap seam to `DISTILL_BACKEND_URL`, currently Qwen3.8-27B. Responses include model identity, while swaps affect no consumer. Thus §6 can vary the model while holding grounding constant, or vary the serving path while holding the model constant.

5 The Copy Ceiling

We treat the copy ceiling as the paper's central contribution, and we therefore provide a formal definition, an explicit list of assumptions, and a dedicated robustness design, in line with the metric-first precedent [9, 30].

Table 2: The request pipeline. Retrieval, gating and the scaffold serialisation need no model; only the final delegation calls one. The single injection authority (match + gate) is the sole path by which any candidate reaches the model, whether lexical or, when enabled, semantic.

Stage	Mechanism	Model?
Match	Lexical inverted index over 8,146 class titles; 2.02 ms p50 → seed classes	no
Gate	Single injection authority: budget scaled to retrieval score, skip below threshold	no
Scaffold	Budget-clamped serialisation of the matched units' markdown-with-ontology	no
SPARQL / search	Read-only, prologue-aware LIMIT-clamped oxigraph over the reasoned closure	no
Semantic fallback	In-process HNSW over 384-d embeddings; fires on lexical miss → candidate seeds into the same gate	*default-off*
Delegate	Inject scaffold into the last user message; call `DISTILL_BACKEND_URL`	yes

5.1 Definition and assumptions

Fix a question q_i with gold target set G_i (the {slug, title} pairs the graph asserts as its answer), the exact scaffold text S_i shown to the model, and answer a_i. Let $\mathrm{m}(t, X) \in \{0, 1\}$ be the deterministic surface matcher, equal to one when the normalised title of gold item t, or at least 80% of its length-≥ 4 words, appears in X. Define the exposed count and per-item **copy ceiling**

$$e_i = \sum_{t \in G_i} \mathrm{m}(t, S_i), \qquad c_i = \frac{e_i}{|G_i|},$$

and the grounded recall $r_i = \left(\sum_{t \in G_i} \mathrm{m}(t, a_i)\right)/|G_i|$. A no-op extractor returning S_i verbatim scores exactly c_i. For the raw arm S_i is empty, so $c_i \approx 0$.

Assumptions (when the instrument applies).

(A1) **Deliberate exposure.** Questions, gold and scaffold derive from one graph, so gold is deliberately present in the injected text. The ceiling measures this exposure; it is not a leak.

(A2) **Shared matcher.** c_i and r_i use the same matcher, making both lower bounds under paraphrase and their difference first-order insensitive to under-counting. Cancellation is approximate because error rates may differ between structured scaffold text and generated prose; where paraphrase is a live threat we re-score with a semantic judge (§6).

(A3) **Per-item, model-free ceiling.** c_i depends only on (q_i, S_i) and is identical across models given a fixed scaffold, enabling comparison across a model sweep.

(A4) **Graph-derived gold.** The ceiling is informative only when gold derives from the injected corpus. Where gold is independently authored (the out-of-domain arm), $c_i \approx 0$ and judged quality carries the axis.

A recall reference point, not an endorsement. Although the no-op extractor scores c_i, it is not an admissible answer: returning the entire scaffold lacks the required precision, conciseness and relevance. The ceiling is therefore a recall reference for delivery fidelity, not a recommendation to reproduce the context.

5.2 Gain over copy

We define the signed **gain over copy** as $g_i = r_i - c_i$, reporting the headline figure $\bar{g} = \bar{r} - \bar{c}$. It re-centres grounded recall against the exposure reference.

- $g \approx 0$, *delivery fidelity*: the model restates nearly everything shown. Recall measures omission rather than fabrication, leaving additions beyond exposed gold unmeasured; for private-knowledge grounding, restatement is the target behaviour.
- $g < 0$, *lossy restatement*: the model drops surfaced gold. The decomposition of §7.2 measures this directly: the negative gain is almost entirely exposed-but-omitted gold (n_{10}), with unexposed recovery (n_{01}) negligible. The magnitude distinguishes models where near-identical raw uplift cannot.
- $g > 0$ would evidence *reasoning over structure*, such as composing an implied but unstated relation. We observe essentially none in-domain ($n_{01} = 3$ of 11,360; §7.2). A positive g alone would not prove reasoning: parametric leakage, guessing or matcher asymmetry could also produce it, and only the multi-hop design of §12 could separate them.

A near-ceiling result supports "faithful delivery" and rules out "reasoning over ontological structure". Figure 2 illustrates the two paths compared.

5.3 Negative controls: what would move the gain

A positive delivery result might reflect lexical overlap rather than knowledge transfer. Four arms hold the model and question fixed while perturbing only the injected block (defined here; results in §7.4):

true the block actually served by the node, injected into the bare model. It should reproduce the live loom arm, showing that the serving path adds nothing beyond the block.

shuffled the same block with body sentences shuffled while retaining the wrapper and seed. Survival of the gain would indicate lexical soup. This weakly probes relational structure because whole-sentence shuffling preserves facts within sentences.

masked the same block with every seed-class title replaced by `Entity-`k, preserving structure but removing

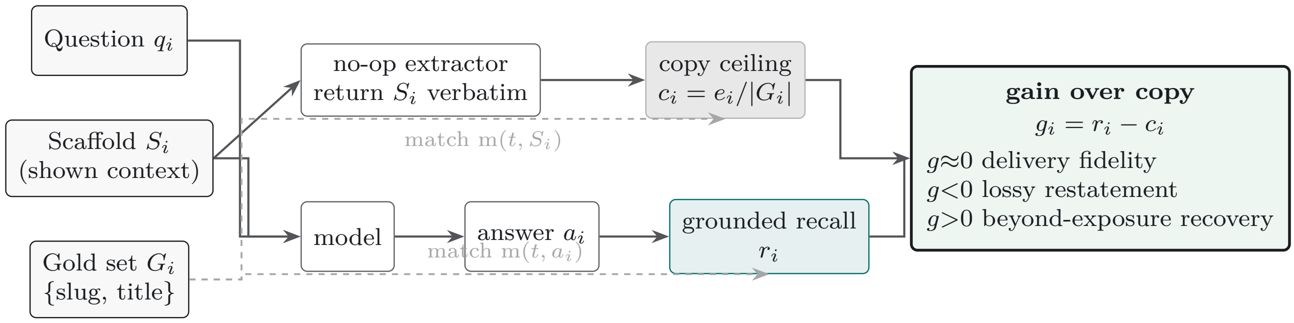

Figure 2: The copy ceiling and the signed gain over copy on one panel. The same shown context S_i feeds two extractors: a no-op extractor that returns S_i verbatim, whose recall against the gold set G_i is the per-item *copy ceiling* c_i; and the model, whose answer a_i scores the grounded recall r_i. Both scores use the identical surface matcher $\mathrm{m}(t, \cdot)$ with G_i as reference (dashed), so their difference, the *gain over copy* $g_i = r_i - c_i$, is first-order insensitive to the matcher's under-counting. The sign reads directly: $g \approx 0$ is faithful delivery (the target behaviour for private-knowledge grounding), $g < 0$ is lossy restatement of surfaced gold, and $g > 0$ would evidence recovery of gold beyond what the exposure check finds (§5).

entity names. Survival would indicate name matching rather than knowledge. Because answer-bearing target names remain, this is a weak probe of name matching.

irrelevant the well-formed, on-corpus scaffold of a *different* question, using a fixed derangement over engaged items. Survival would show that any ontology-shaped text suffices.

If gain over copy remains near zero under **shuffled**, **masked** and **irrelevant**, the ceiling tracks exposure and delivery is content-bound; a shift indicates leakage through surface form or position. This experiment tests whether the observed lift is content-specific, serving as the direct analogue of a shortcut-robustness check for a faithfulness metric [9].

6 Method

Design. Both studies use within-model pairs: the *same* model answers the *same* question twice, cancelling between-question difficulty variance and yielding a paired per-question estimand. The ten-model sweep varies the *model* with grounding fixed; the new study varies the *serving path* with the model fixed.

Study 1: the ten-model sweep. The knowledge graph supplies templated questions for every sufficiently-defined class: **T-REL** (requirements, uses, dependencies, enablers, and parts), **T-TAX** (immediate and ancestral parents), and **T-COMMON** (shared ancestors of a pair). Gold answers are the {slug, title} targets asserted by the graph. Seed 42 yields **510 questions** across 15 domains: 285 T-REL, 180 T-TAX, and 45 T-COMMON. Each is posed *raw* and *scaffold* (with a budget-clamped 1,500-token ontology extract prepended) to ten models from five providers under one deterministic configuration (`temperature`=0, `max_tokens`=2048; `reasoning_effort`=low only where thinking cannot be disabled, preventing mandatory passes from truncating answers). The lexical matcher from §5 counts a gold item when its normalised title, or $\geq 80\%$ of its long words, appears in the answer. Because this under-counts paraphrase, absolute recall is a lower bound; the signal is the explicitly lexical *paired* delta for the same question, scorer, and model. We detected truncation per row: GLM-4.6 truncated on 174 rows, of which 5 were retried and the remainder scored as received, making that row lower-confidence; Gemini 2.5 Flash-Lite truncated on 16.

Study 2: the production-node paired study. We hold the model constant at Qwen3.8-27B on the reference GPU host, isolating the serving path. The *loom* arm calls the production Rust node's `/v1/chat/completions` endpoint, which retrieves, confidence-gates and injects the scaffold, returning telemetry (engagement, injected tokens, fusion path) in the response `loom` block. The *raw* arm calls the same backend without a scaffold. Both use `temperature`=0 and `max_tokens`=1536, below which the reasoning backend truncates output to empty. Three pre-generation frozen sets contain **117** questions: 24 *arcane* (deep in-domain), 33 *thin* (sparsely covered), and 60 *general* (out-of-domain). The general set is also graded by the out-of-domain judged arm below, keeping the judged arms comparable. The four negative-control arms from §5.3 use the same node. Their scaffolds come from the LLM-free `/loom/scaffold` endpoint, ensuring that each control presents the bare model, via a system message, with exactly the block served to the loom arm.

Out-of-domain judged arm. A five-model judged arm poses the 60-question *general* set (adjacent, in-domain-general and off-domain subsets) to *harnessed* and *bare* configurations. Each pair is graded 0–5 against frontier-authored key points, providing an independent out-of-domain instrument (§7.5).

Cross-family semantic judge. To cover paraphrase across Study 2's in-domain and out-of-domain questions, we use a reference-guided **0–5 rubric** judge rather than lexical scoring. The judge is `openai/gpt-4.1` via OpenRouter with `temperature`=0 and `max_tokens`=200, applying the same fixed system-message rubric in every arm. It is from a different family than the Qwen candidate, limiting self-preference [33, 42]; it is blind to each answer's arm and study origin, and grading can resume per (set, id, arm). References are graph-derived targets in-domain and frontier-authored key points out-of-domain.

Statistics. Per set and pooled, the primary metric is the paired mean difference with a seeded **10,000-resample bootstrap** 95% percentile interval [7, 20]. We also report a two-sided **Wilcoxon signed-rank** test and the paired effect size, **matched-pairs rank-biserial correlation** $r = (T^+ - T^-)/(T^+ + T^-)$, over non-tied pairs. This is preferable to Cliff's delta for matched pairs because Cliff's independent-samples dominance statistic divides the win−loss margin by the many ties. Set-level p-values receive a **Holm–Bonferroni** correction [5]. For out-of-domain non-regression, an *equivalence* claim, we use two one-sided tests (TOST) with a pre-specified ±0.25 margin [22], rather than treating a non-significant difference as no effect. Because in-domain graph-derived questions cluster by class and domain [5], we report a domain-clustered block bootstrap alongside the naive interval and an intention-to-treat delta retaining questions where the gate did not engage [17].

7 Results

Reporting deviations (Study 2). Two deviations affect interpretation of the production-node figures. *(i) Budget-exhaustion re-runs.* At `max_tokens`=1536, 42 of 234 loom/raw completions (117 questions × two arms) exhausted the budget during reasoning and produced no output. Each affected pair was re-executed at 4096 in *both* arms; three questions remained empty and were excluded, yielding a complete-case $n = 114$ paired analysis (thin: 33 to 30). The pooled contrast is therefore a mixed-budget estimand: 39 surviving re-run pairs use 4096 and the remainder 1536. *(ii) Controls not retried.* The four negative-control arms remained at 1536 and had substantial empty attrition: true 24/56 (42.9%), shuffled 20/56 (35.7%), masked 23/56 (41.1%), and irrelevant 28/56 (50.0%). Control contrasts are consequently lower-confidence, complete-case estimates with small, unequal n. Attrition also indicates a scaffold-length × reasoning-budget interaction: empty rates increased with scaffold length and disorder. The live loom arm's original-1536 empties were 2/24 arcane and 19/33 thin, versus raw 6/24 and 15/33; we revisit this in §12.

7.1 In-domain: delivery fidelity

For Gemini 3.7 Flash, raw recall is 0.375 and scaffold recall 0.942, a paired delta of +**0.567** with naive and domain-clustered 95% CIs of [+0.532, +0.603] and [+0.524, +0.611], respectively, across $n = 510$ items; there were no errors or truncated responses (Figure 3). The low raw score is consistent with a private domain largely absent from pretraining. The copy ceiling is **0.964**, placing recall −0.022 below the fraction of gold already exposed. Thus the model returns most curated answers but omits some; because recall does not measure additions beyond the gold, fabrication and attribution precision remain unmeasured (§11). §7.2 tests whether models recover lexically unexposed gold or drop exposed facts.

7.2 Ten models: gain over copy

Table 3 applies the paired design to ten models from five providers under one deterministic configuration. Raw→scaffold uplift ranges from +0.57 to +0.78, while scaffold recall spans 0.897–0.942 against the model-independent ceiling of **0.964**. Hence raw uplift does not distinguish delivery from exposure.

The signed **gain over copy** is negative throughout (Figure 4), from −0.067 for Gemini 2.5 Flash-Lite to −0.022 for Gemini 3.7 Flash. Because $g_m = R_m - 0.964$, ranking by gain equals ranking by scaffold recall; its contribution is an interpretable scale, not a new ordering. Recent models sit nearest the ceiling, while Gemini 2.5 Flash-Lite, GPT-4.1-mini and Mistral-Small-24B fall further below, an ordering specific to this task. Two models had transport-level truncations under the fixed 2048-token budget: GLM-4.6 had 174 truncated and 5 retried, with the rest scored as received and its row treated as lower-confidence; Gemini 2.5 Flash-Lite had 16 truncated.

What adds information: the exposure/recovery decomposition. A negative g alone cannot distinguish omission of exposed gold from failure to recover unexposed gold. We therefore classify each item by scaffold exposure and answer recovery. Across ten models and 11,360 gold items, n_{11} = 9,865 were exposed and recovered, n_{10} = 735 exposed and omitted, n_{01} = 3 unexposed and recovered, and n_{00} = 757 unexposed and omitted (Table 4). Thus only **3 of 11,360** items (0.026%) were recovered without exposure, with no model exceeding $n_{01} = 1$. With $n_{01} \approx 0$, $g \approx -n_{10}/|G|$: the deficit chiefly measures imperfect copying, including lexical-match under-counting of paraphrase, rather than reasoning. Context utilisation, $n_{11}/(n_{11} + n_{10})$, ranges from 0.900 to 0.955, implying roughly one exposed item in fourteen is dropped. It is micro-averaged over exposed items, whereas recall averages per-question ratios with varying c_i; accordingly, the rankings differ once: llama-3.3-70b has utilisation 0.934 versus deepseek-chat's 0.931, but recall 0.916 versus 0.924.

7.3 The production-node paired study

Study 2 holds Qwen3.8-27B and decoding constant on the reference host while varying only whether requests pass through the shipped Loom node (*loom*) or reach

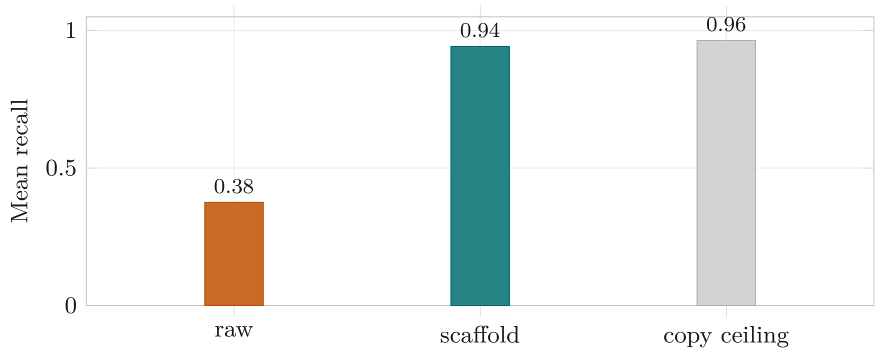

Figure 3: Gemini 3.7 Flash, $n = 510$. The scaffold arm (0.942) lands just under the *copy ceiling* (0.964), the recall a no-op extractor of the injected context would already achieve. The +0.57 raw→scaffold uplift is almost entirely answer exposure, not reasoning over structure.

Table 3: Ten models, one deterministic configuration, $n = 510$ (seed 42), sorted by gain over copy (scaffold recall − copy ceiling). The copy ceiling (0.964) is a property of the questions and the injected context and is identical across models. The interval shown is the domain-clustered block bootstrap (the wider, primary one).

Model	Raw	Scaffold	Uplift [clustered 95% CI]	Ceiling	Gain
Gemini 3.7 Flash	0.375	0.942	$+0.567$ $[+0.524, +0.611]$	0.964	-0.022
Gemini 3.5 Flash-Lite	0.323	0.938	$+0.615$ $[+0.565, +0.667]$	0.964	-0.026
Claude Haiku 4.5	0.151	0.934	$+0.783$ $[+0.741, +0.827]$	0.964	-0.031
GLM-4.6	0.243	0.928	$+0.685$ $[+0.641, +0.727]$	0.964	-0.036
DeepSeek-Chat	0.347	0.924	$+0.578$ $[+0.532, +0.622]$	0.964	-0.040
Llama-3.3-70B	0.227	0.916	$+0.688$ $[+0.632, +0.738]$	0.964	-0.049
Qwen-2.5-72B	0.309	0.914	$+0.605$ $[+0.562, +0.649]$	0.964	-0.050
Mistral-Small-24B	0.285	0.906	$+0.621$ $[+0.558, +0.682]$	0.964	-0.058
GPT-4.1-mini	0.162	0.905	$+0.743$ $[+0.697, +0.787]$	0.964	-0.060
Gemini 2.5 Flash-Lite	0.227	0.897	$+0.670$ $[+0.623, +0.716]$	0.964	-0.067

the backend directly (*raw*). The node includes live retrieval, confidence gating and injection authority; a cross-family GPT judge scores answers 0–5 as described in §6. Table 5 reports paired results.

Across $n = 114$, loom raises judged quality by $+\mathbf{0.27}$ points ($[+0.11, +0.45]$, Wilcoxon $p = 0.0036$, matched-pairs rank-biserial $r = +0.589$), with 24 wins, 8 losses and 82 ties. The arcane-set effect is $+\mathbf{0.79}$ ($[+0.17, +1.42]$, $p = 0.033$, Holm-adjusted $p = 0.099$, $r = +0.670$); the thin-set estimate is $+0.30$ ($[-0.03, +0.63]$), positive but non-significant. For general questions, $+0.05$ $[0.00, +0.13]$ lies within the pre-specified ± 0.25 equivalence margin, establishing non-regression by CI inclusion (§7.5). The gradient is consistent with grounding and inconsistent with a uniform prompt-formatting effect (Figure 5).

Injection engagement is 95.8% on arcane, 100% on thin and 78.3% on general. Thus the general-set null cannot be explained chiefly by the gate skipping questions, although it skipped the remaining 21.7%. Median injected context is ≈ 1,200–1,300 tokens across sets. Median latency is comparable for arcane, 89.8 s (loom) versus 93.4 s (raw), and general, 66.0 versus 65.7 s. Thin is slower under loom, 172.4 versus 145.2 s, consistent with full engagement and longer scaffolds.

7.4 Negative controls: attributing the gain

Four controls test whether the live lift reflects knowledge transfer or merely lexical overlap with an injected block. Each holds model and question fixed while perturbing only the scaffold fetched verbatim from the LLM-free `/loom/scaffold` endpoint (§5.3). Table 6 and Figure 6 report lower-confidence, complete-case contrasts pooled across arcane and thin.

The exact block passed directly to the bare model reproduces the live arm: loom − true is -0.06 $[-0.38, +0.25]$, indicating no detectable contribution beyond injection. True − raw is $+0.59$ $[+0.16, +1.06]$ ($p = 0.026$). By contrast, an on-corpus scaffold from another question gives irrelevant − raw $+0.04$ $[-0.54, +0.57]$ ($p = 0.73$). Its near-zero point estimate is consistent with no generic benefit from ontology-shaped text, but attrition leaves the interval too wide to establish equivalence. True − irrelevant is $+0.35$ $[0.00, +0.78]$ ($p = 0.091$, $n = 23$), consistent with content-specific transfer but imprecise and not conventionally significant.

The placebo had to be verified irrelevant. An initial shift-by-one derangement kept donor and target within the same domain cluster, preserving vocabulary and neighbouring concepts. We therefore imposed seed-IRI disjointness, verifying that each donor shared

Table 4: Exposure/recovery decomposition over the 510-question sweep (1,136 gold items per model; 11,360 pooled), sorted by context utilisation. Utilisation is $n_{11}/(n_{11}+n_{10})$, the fraction of *exposed* gold the answer recovers; n_{10} is exposed gold the answer omitted; n_{01} is gold recovered though the scaffold did not expose it. Across all ten models n_{01} never exceeds 1, and is 3 pooled: recovery beyond exposure is empirically negligible. Counts are pooled over gold items (micro-averaged), whereas the headline recall and ceiling average per-question ratios (macro-averaged), so these counts do not reconstruct those numbers directly.

Model	Context utilisation	n_{10} (exposed, omitted)	n_{01} (unexposed, recovered)
Gemini 3.7 Flash	0.955	48	0
Gemini 3.5 Flash-Lite	0.947	56	0
Claude Haiku 4.5	0.946	57	1
GLM-4.6	0.945	58	0
Llama-3.3-70B	0.934	70	0
DeepSeek-Chat	0.931	73	0
Qwen-2.5-72B	0.924	81	1
Mistral-Small-24B	0.917	88	1
GPT-4.1-mini	0.908	98	0
Gemini 2.5 Flash-Lite	0.900	106	0
Pooled	**0.931**	**735**	**3**

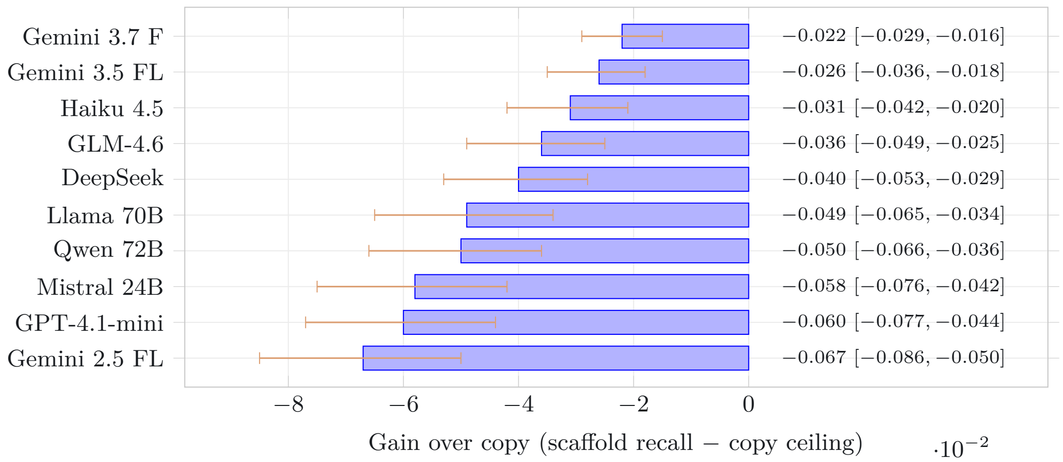

Figure 4: Signed gain over copy for all ten models, negative throughout, with per-model seeded 10,000-resample bootstrap 95% intervals (whiskers); the value column at right gives each point and interval. No model's interval crosses zero, and none exceeds the copy ceiling on the lexical metric; the spread (−0.067 to −0.022) is the axis that separates them where the near-identical raw uplift does not. Models nearest zero (top) restate the vetted facts most faithfully.

neither a seed IRI nor seed class with the target gold. Only then did the point estimate fall to ≈ 0. Unverified irrelevance would have contaminated the placebo baseline.

Masking each seed-class *name* as `Entity-`k yields $+0.42$ over raw; sentence shuffling yields $+0.25$ (both non-significant with wide CIs). These weak probes preserve, respectively, answer-bearing target names and sentence-level facts, so they bound surface-form contributions only loosely. Common-pair estimates are true − shuffled $+0.30$ $[-0.10, +0.73]$ ($n = 30$) and true − masked $+0.15$ $[-0.19, +0.44]$ ($n = 27$), both non-significant. The curation-depth gradient is clearer: true − raw is $+0.95$ $[+0.37, +1.63]$ ($p = 0.018$) on arcane and $+0.08$ $[-0.46, +0.62]$ (ns) on thin, independently matching the live study.

7.5 Out-of-domain: the gate holds

In the five-model judged arm (§6), a cross-family judge scored 60 out-of-domain questions 0–5 against frontier gold. Every harness−bare 95% interval includes zero: all questions -0.05 $[-0.11, 0.00]$, off-domain -0.09 $[-0.25, +0.02]$, and the pre-specified worst case of erroneous injection -0.23 $[-0.63, +0.05]$ (Table 7). The gate correctly skipped 60 of 100 off-domain gradings, making harnessed and bare answers identical; detected degradation was confined to over-firing and concentrated on the weakest model. Study 2 provides the stronger equivalence result: general-set $\Delta = +0.05$ $[0.00, +0.13]$ lies within the pre-specified ± 0.25 TOST margin. These results bound regression near zero but do not prove zero effect.

8 Analysis: Exposure, not Reasoning

Both studies indicate that graph-grounding uplift over a private corpus is almost entirely *exposure*: faithful delivery of curated facts shown to the model, rather

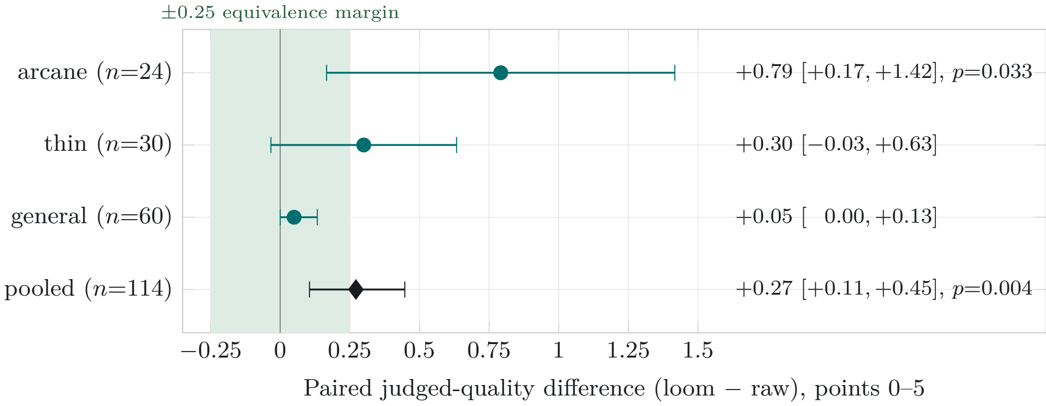

Figure 5: Study 2 paired judged-quality deltas (Qwen3.8-27B, model held constant, loom versus raw serving path; cross-family GPT judge), per set and pooled, with seeded 10,000-resample bootstrap 95% intervals. The three sets are ordered by curation depth: the effect is largest on the deep *arcane* set (+0.79), attenuates on the sparsely-covered *thin* set (+0.30), and is negligible out-of-domain on *general* (+0.05). That monotone gradient, large where curation is deep and vanishing where the corpus is thin or the question off-domain, is the signature a genuine grounding effect should show and not what a uniform prompt-formatting artefact would produce. The shaded band is the pre-specified ±0.25 TOST equivalence margin: the general-set interval [0.00, +0.13] sits wholly inside it, so out-of-domain non-regression is read by CI-inclusion, not from a non-significant difference (§7.3, §7.5).

Table 5: Study 2, the production-node paired study: Qwen3.8-27B, model held constant, loom versus raw serving path, judged 0–5 by a cross-family GPT judge. Paired mean difference with seeded 10,000-resample bootstrap 95% CI, two-sided Wilcoxon p (Holm-adjusted across the three sets), the matched-pairs rank-biserial r (paired effect size, conditioned on non-tied pairs), and win/loss/tie counts. Complete-case after budget-exhaustion re-runs (§7).

Set	n	Δ (loom−raw)	95% CI	p (Holm)	r	W/L/T
arcane	24	+0.79	[+0.17, +1.42]	0.033 (0.099)	+0.670	10/3/11
thin	30	+0.30	[−0.03, +0.63]	0.118 (0.237)	+0.431	12/5/13
general	60	+0.05	[0.00, +0.13]	0.180 (0.237)	+1.000†	2/0/58
pooled	114	+0.27	[+0.11, +0.45]	**0.0036**	+0.589	24/8/82

† The general-set r rests on 2 non-tied pairs (58 ties); read it as directional only, not as a magnitude.

than reasoning over injected structure. The instrument establishes this distinction and yields two engineering implications.

The headline is delivery fidelity, and the instrument is what reveals it. In the ten-model sweep, raw uplift is large and consistent (+0.57 to +0.78), whereas gain over copy is small and uniformly negative (−0.067 to −0.022). Every model appears transformed against its bare-model baseline, but none exceeds a verbatim copy of its context; they differ only in how much exposed gold they drop. Thus uplift reveals little about model performance, while gain over copy discriminates between models. This inversion supports reporting the ceiling as standard practice.

The live controls point to content, not form. The served block raises judged quality (true−raw +0.59), while a same-shaped, on-corpus but wrong block has a near-zero point estimate (irrelevant−raw +0.04). The content-specific residual is +0.35; masking names (+0.42) or shuffling sentences (+0.25) costs comparatively little. The graded facts therefore appear to carry most of the effect, although the arms do not precisely isolate names and order. The arms use complete cases from unequal survivor sets, their contrasts consequently do not sum arithmetically, and several are non-significant with wide intervals. Nonetheless, a content-specific lift alongside a verified-irrelevant placebo near zero fits faithful exposure better than generic priming by ontology-shaped text. Crucially, this pattern emerged only after verifying that the irrelevant donor was seed-IRI-disjoint: an assumed-inert placebo had leaked domain vocabulary and produced a spurious baseline.

Consequence 1: the copy ceiling is a regime detector. Compute it before you architect. Calculate the ceiling on representative questions before selecting an architecture. A ceiling close to 1 (0.964 here) indicates that answer entity names are exposed in the context, although it does not prove that the required relation is present or understood; the following regimes are therefore heuristic. A high ceiling identifies a *serving regime*: retrieval supplies most target coverage, while generative re-encoding may lose fidelity, though generation can still improve selection, relevance and presentation. The target is faithful delivery, measured by gain over copy. A low ceiling identifies a *synthesis regime*, in which answers must be

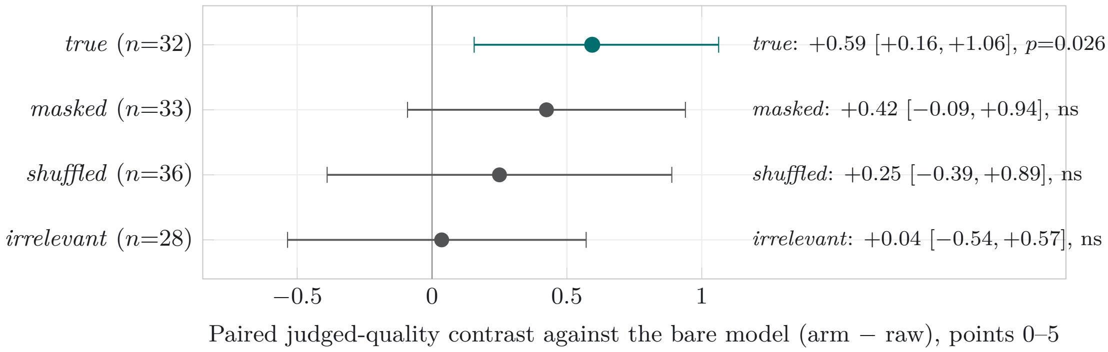

Figure 6: Negative-control decomposition (Study 2, pooled arcane + thin, judged 0–5 by the cross-family GPT judge), each arm holding the model and question fixed and perturbing only the injected block, read as a paired contrast against the bare model (arm − raw) with seeded bootstrap 95% intervals. Read top to bottom the ladder is the paper's central negative control: the served block lifts quality (*true* − raw +0.59, the one significant contrast, in teal), masking the seed-class names (+0.42) or shuffling the body sentences (+0.25) costs comparatively little, and an equally well-formed, on-corpus scaffold drawn from a *different* question shows a near-zero point estimate (*irrelevant* − raw +0.04). That near-zero placebo is consistent with the lift being specific to scaffold *content* rather than the mere presence of ontology-shaped text, though its wide interval under attrition means inertness is not demonstrated by equivalence; the content-specific residual is *true* − *irrelevant* = +0.35 [0.00, +0.78]. The arms are complete-case over unequal survivor sets and all but *true* are non-significant with wide intervals, so the attribution rests on the pattern across arms, not any single contrast (§7.4).

Table 6: Negative-control contrasts, pooled arcane + thin, judged 0–5, paired mean difference with seeded bootstrap 95% CI and two-sided Wilcoxon p. Complete-case over surviving pairs; the control arms were not retried at the larger budget, so n varies by arm (§7). Read top to bottom: the serving path adds nothing (loom−true ≈ 0); the served block lifts quality (true−raw); the same-shaped placebo does not (irrelevant−raw ≈ 0); the content-specific residual is true−irrelevant.

Contrast	n	Δ	95% CI	Wilcoxon p
loom − true	32	−0.06	[−0.38, +0.25]	0.81
true − raw	32	+0.59	[+0.16, +1.06]	0.026
masked − raw	33	+0.42	[−0.09, +0.94]	0.15
shuffled − raw	36	+0.25	[−0.39, +0.89]	0.42
irrelevant − raw	28	+0.04	[−0.54, +0.57]	0.73
true − irrelevant	23	+0.35	[0.00, +0.78]	0.091
true − shuffled	30	+0.30	[−0.10, +0.73]	0.209
true − masked	27	+0.15	[−0.19, +0.44]	0.398
Per set (true − raw): arcane +0.95 [+0.37, +1.63], $p = 0.018$; thin +0.08 [−0.46, +0.62], ns.				

composed across retrieved pieces and GraphRAG-style machinery (community summaries, multi-hop traversal, GNN soft-prompts) may add value. That value must be measured above the ceiling rather than against a bare-model baseline. In a high-ceiling setting, bare-model uplift is delivery masquerading as reasoning. We suspect much reported graph-RAG gain reflects undiagnosed serving-regime exposure; the ceiling distinguishes the regimes before expensive architecture is built.

Consequence 2: value concentrates at curation, and the instrument doubles as a promotion gate. When the model delivers answers supplied by the corpus, investment should favour upstream curation, whose cost is amortised rather than repaid per query. Automated extraction candidates, such as blocks proposed by OntoCast [13] (analysed here solely as a third-party upstream producer), can be evaluated through the copy ceiling of the questions they answer. A block that drives the ceiling towards 1 is answer-complete and eligible for promotion; one that does not is unready. Thus the serving-layer audit also grades ingestion, with a block's answer-completeness approximated by its copy ceiling. This governance gate is necessary but insufficient: term-stuffing answer names can inflate the ceiling while leaving a block incoherent or wrong. Exposure coverage must therefore complement existing CI, consistency and human-review gates.

The honest discovery arc. We built the serving node, observed substantial uplift, and questioned whether it merely reproduced GraphRAG with better provenance. The copy ceiling posed the adversarial question: *is any of this reasoning, or is it all exposure?* Uniformly negative gain over copy across ten models, together with a placebo point estimate that collapsed once verified irrelevant, answers: exposure. For a private-knowledge product, this meets the specification: vetted sources provide trust, named generation provides attribution, and one-time curation controls cost. The contribution is the instrument that distin-

Table 7: Out-of-domain non-regression (five-model judged arm): paired harness−bare quality delta (judge 0–5 vs frontier gold), five models, seeded paired bootstrap 95% CI. Every interval includes zero. The worst case is off-domain questions where the confidence gate mis-fired and injected anyway; the last row is where it correctly skipped.

Subset	Mean Δ	95% CI	n pairs
All general questions	−0.05	[−0.11, 0.00]	300
adjacent	+0.01	[−0.02, 0.05]	100
in-domain-general	−0.06	[−0.14, 0.00]	100
off-domain	−0.09	[−0.25, 0.02]	100
Off-domain, gate mis-fired (worst)	−0.23	[−0.63, 0.05]	40
Off-domain, gate skipped (correct)	+0.00	n/a	60

guishes these effects and the discipline of reporting it.

9 A Production Case Study: Podcast Assertion Extraction

A production deployment tested whether scaffold grounding helps a model *write into* the corpus. The node's podcast-ingest pipeline extracts tiered, source-attributed assertions from episode transcripts; each carries `ontology_terms`, concept labels used to locate a destination page. Assertions whose terms resolve to no page are discarded or queued for page proposal, making *term resolution against the live graph* the landing metric. We report a lexical proxy: normalised string matches between emitted terms and the inventory of 8,142 classes. This is not a human judgement of assertion quality.

We ran the unmodified production prompt over ten episodes of a technology news podcast (November 2025–August 2026) across ten arms: the node's local Qwen3.8-27B *through the Loom* (scaffold injected at the default 1,500-token budget, verbatim short-circuit disabled) and *bare*; and eight frontier or near-frontier cloud models, all bare, because the Loom grounds only the model behind its own façade. All arms used direct chat-completion calls with identical sampling (temperature 0.2, 12,288 max tokens). Claude and GPT used a model router; the others used their providers' APIs. One Loom call produced three assertions for one episode and remains in the totals.

The paired scaffold effect is large: with the same weights, prompt and GPU, term resolution rises from 0.17 to 0.55 (3.2×) while assertion count falls. Grounding narrows the model's vocabulary towards the graph's, as integration requires. Generation time also roughly halves (173s vs. 346s per episode), consistent with the scaffold displacing open-ended reasoning. The grounded local 27B exceeds every bare frontier arm (best: 0.42), reproducing Study B's read-path pattern in a write path. This does not establish that frontier models are worse extractors: several produce more assertions, and assertion quality is not judged here. Without the graph's vocabulary, however, their concept labels miss it and the assertions do not land. The pipeline also depends on fidelity to noisy input. Several arms reproduce speech-recognition errors as entity names (7/181 for GLM-4.7; 3 each for DeepSeek and Opus 4.8), whereas both local Qwen arms normalise them. Corrupted names damage page matching and downstream dedup fingerprints. Figure 7 shows all three observations on one panel.

The prior production default disabled scaffold injection to work around the façade's verbatim short-circuit, which answered transcript-shaped prompts from the scaffold without invoking the model. The pipeline now requests injection while explicitly declining that short-circuit. This remains *single-configuration estimation*: one podcast, ten episodes, one run per arm and one lexical scorer yield a point estimate from one draw of one setup, with no sampling variance or transfer test across corpora or scorers. The paired Loom-versus-bare contrast is least exposed because both arms share the draw; another draw could more plausibly reorder the cross-model rows. This is evidence about where extracted knowledge lands, not a model-quality leaderboard.

9.1 The judged page: an end-to-end quality gate

Term resolution does not show whether a page improves. For each arm, we therefore applied its matched assertions to sandbox copies of target pages using the production integration mechanic. A cross-family judge (Gemini 3.1 Pro, temperature 0, blind A/B with seeded order randomisation, un-blinded at analysis) scored each before/after pair. A second judge, GPT-5.6, rescored a 20% subset with 93% verdict agreement. Every arm degrades judged page quality (means −1.04 to −0.44); the two arms adjacent to a judge's family are flagged in the released analysis.

The insertion mechanic does not alone explain the result. A section-splice rewriter emits only an anchored section edit, which deterministic code applies fail-closed while preserving the original by construction. Prompt optimisation covered four variants on a dev split, followed by validation on held-out pages under two rubrics. One rubric was written post-hoc, disclosed as such, to credit informativeness explicitly. The best variant remained negative under both judges (−0.40 dev, −0.90 held-out). Persistence across mechanic, rubric and judge suggests a structural problem: narrow, newly extracted facts dilute a mature curated page.

Table 8: Assertion extraction across ten serving arms, ten episodes. *Ground* is the fraction of emitted `ontology_terms` that resolve to an existing knowledge-graph page (lexical proxy); *Artif.* counts claims that reproduce speech-recognition artefacts as entity names (e.g. "Opus 48", "GPT55"). n is total assertions; Conf. is the model's own mean stated confidence, which is self-reported and not comparable as a quality measure across families.

Arm	n	Conf.	Ground	Artif.	s/episode
Loom + Qwen3.8-27B (local)	125	0.83	**0.55**	0	173
Qwen3.8-27B bare (local)	170	0.81	0.17	1	346
Gemini 3 Flash (preview)	107	0.80	0.42	1	13
GLM-4.7	181	0.80	0.39	7	17
DeepSeek v4 Flash	131	0.86	0.28	3	66
Claude Opus 4.8	171	0.75	0.27	3	51
Claude Haiku 4.5	295	0.82	0.22	0	242
GPT-5.6	133	0.82	0.21	0	23
GLM-5.3	169	0.82	0.21	0	33
Claude Sonnet 5	175	0.76	0.20	0	50

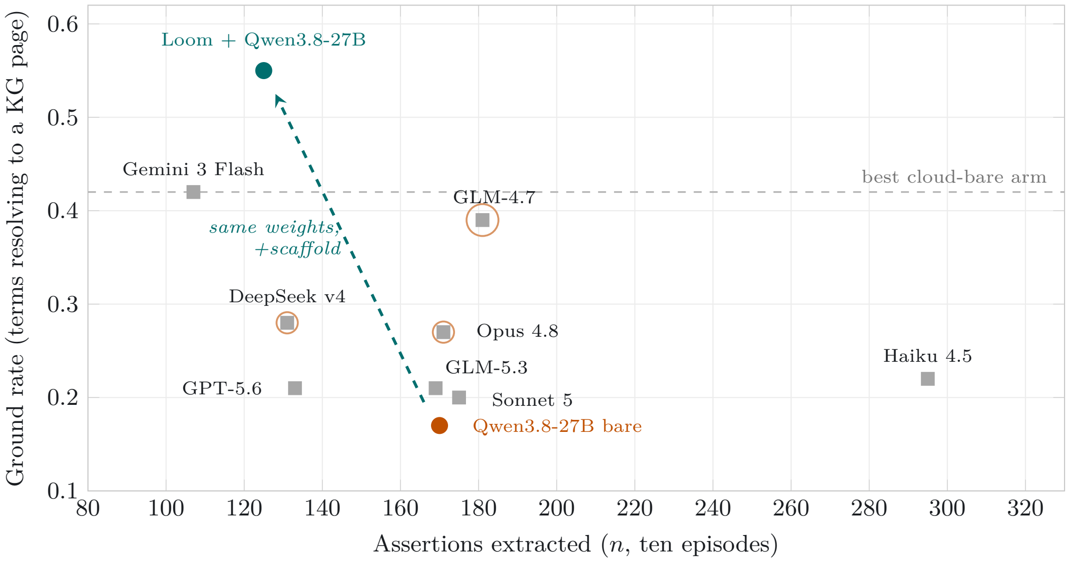

Figure 7: The case study on one panel (Table 8). Each mark is a serving arm: volume of extracted assertions (n) against the rate at which its `ontology_terms` resolve to knowledge-graph pages. Grey squares are cloud models run bare; a burnt-orange halo marks arms with more than two speech-recognition artefacts copied into claims (radius scales with count). The dashed teal arrow is the paired contrast: identical local weights move from 0.17 to 0.55 when served through the Loom, clearing the entire bare field, whose best arm (dashed grey line, 0.42) no volume of extraction reaches.

Under a pre-declared decision rule, the integration phase was disabled while extraction and verification continue. Redesign will address *placement*, towards thin pages, dedicated currency sections or batched section regeneration, rather than further prompt iteration. The instrument earned its keep in the direction the promotion-gate discipline of §8 predicts: it caught a value-destroying step before it ran at scale.

10 Two-Edge Composition: Locating the Synthesis Boundary

The exposure/recovery decomposition found beyond-exposure recovery in only 3 of 11,360 gold items, but did not test whether models *would* compose fully exposed ontological structure when required. The multi-hop design sketched in §12 does. We mined two-edge chains $A \xrightarrow{r_1} B \xrightarrow{r_2} C$ over the restriction-encoded relations *requires* and *hasPart*. Eligible chains had no direct asserted or inferred $A \to C$ edge, omitted C from A's served block, and had at most three valid targets. Of 6,514 qualifying chains, 150 were sampled across the four relation-pair shapes. Questions name only A ("Name a part of something A requires"). Every arm serves exactly four seed-shuffled blocks, so presence does not signal relevance. The six arms provide *both* premises (A's and B's blocks plus two seed-disjoint distractors), either premise *alone* (plus three distractors), *masked* premises (both with relation labels replaced), *irrelevant* context (four distractors), or *raw* input (no context). Twelve models ran all arms under identical sampling.

The controls behave as required: a-only, irrelevant and raw recall are at floor for every model (zero in almost all cells; none exceeds 2/150), showing no parametric leakage or placebo effect. The contrast $\Delta = \text{both} - b\text{-only}$ isolates composition from in-context guessing because b-only exposes the target verbatim but withholds the route. Figure 8 shows three groups, not a capability ladder. Five models compose (positive bootstrap CI): Gemini 3 Flash ($\Delta = +0.33$), the local Qwen3.8-27B (+0.29), DeepSeek v4 (+0.26), Sonnet 5 (+0.25) and Opus 4.8 (+0.20). Three are indistinguishable from guessing, including GPT-5.6, whose b-only recall (0.89), the field's best guess rate, matches its both-premises recall (0.88). Four are *anti-composers*, with significantly negative Δ (GPT-4.1-mini −0.47, Mistral-24B −0.29, GLM-4.7 −0.18, Llama-70B −0.12): adding the route to the answer-bearing block reduces performance. Masking relation labels collapses every model (best masked recall 0.61, most below 0.2), so composition depends on labels rather than bare graph structure.

Three conclusions follow. First, the serving boundary is crossable: whereas gain over copy is uniformly negative in the delivery setting, genuine synthesis separates models decisively, and the local model behind this node's façade leads the composing class. Second, guessing from the answer-bearing block alone is substantial (0.31–0.89 recall) and would masquerade as multi-hop reasoning without single-premise controls; uncontrolled two-hop benchmarks therefore overstate composition. Third, 99.97% of naively mined two-hop chains were unusable because the intermediate class's served block already names the target. With per-IRI serving, most "multi-hop" questions are single-hop exposure in disguise, extending the copy-ceiling lesson one edge outward.

11 The Multivariate Bar

Table 9 specifies the four-axis deployable bar from §2, its instruments and current status. Two axes are measured; two are specified but unresolved. The bar is conjunctive, not reducible to one number.

Axes 1 and 2 establish excellent, attributable delivery where the corpus is authoritative, with no detected regression elsewhere. Separate instruments prevent this conjunction from being a single-metric artefact. Axis 3 requires a web-grounded arm on the same questions, since the bare model is not the correct baseline; axis 4 requires reported attribution precision. Both remain unmeasured, so the full bar is not cleared. This qualification prevents the in-domain result from being mistaken for the entire claim, the failure mode that motivates the copy ceiling.

12 Limitations and Threats to Validity

- **Single corpus.** Both studies use one curated ontology in one broad domain. Although the copy ceiling and decomposition are corpus-general, their magnitudes (a 0.964 ceiling and +0.27 live lift) do not transfer.
- **Single-family judge.** Study 2 uses one GPT-family judge (`openai/gpt-4.1`). It is cross-family from the Qwen candidate, avoiding self-preference, but there is no panel, answer-order randomisation, or repeatability test across judge seeds. Gold references mitigate but do not eliminate judge idiosyncrasy.
- **Complete-case control attrition.** The control arms were not retried at the larger budget and have 36–50% empty attrition at 1536. All contrasts therefore compare unequal survivor sets. Attrition rises with scaffold length and disorder, so survivors are not a random subsample and causal interpretation is unsafe. A uniform-budget rerun at the larger budget is needed. The arms do not sum arithmetically; several contrasts are non-significant with wide intervals (shuffled−raw, masked−raw, true−irrelevant), so attribution rests on the *pattern* across arms.
- **Small per-set n for controls.** Per-set contrasts have $n = 8$–21. The curation-depth gradient (arcane true−raw +0.95 significant, thin +0.08 ns) is suggestive, while thin-set nulls may reflect either low power or no effect.
- **Lexical scorer for Study 1.** The ten-model gain-over-copy and in-domain recall use a lexical title matcher that under-counts paraphrase. Absolute recall and gain over copy are therefore lower-bound estimands. Only Study 2 uses the semantic judge.
- **Semantic fallback disabled throughout.** HNSW semantic retrieval was disabled because its document-embedding recall (0.816) falls below the 0.87 design floor, a red gate retained and reported. The loom arm therefore uses lexical retrieval only. A query-shaped embedding that clears the floor could alter engagement and thin-set results.
- **Single production model.** Study 2 fixes the model at Qwen3.8-27B, isolating the serving-path effect but leaving its magnitude under other backends unmeasured. The ten-model sweep varies models only on the offline scaffold, not the live node.
- **Three excluded questions.** Three questions empty at 4096 in both arms are excluded (complete-case $n = 114$). Although symmetric, this departs from intention-to-treat.
- **Two axes of the bar unmeasured.** The specified web-search baseline (axis 3) and attribution precision (axis 4) remain unmeasured, so the deployable claim is not fully closed.

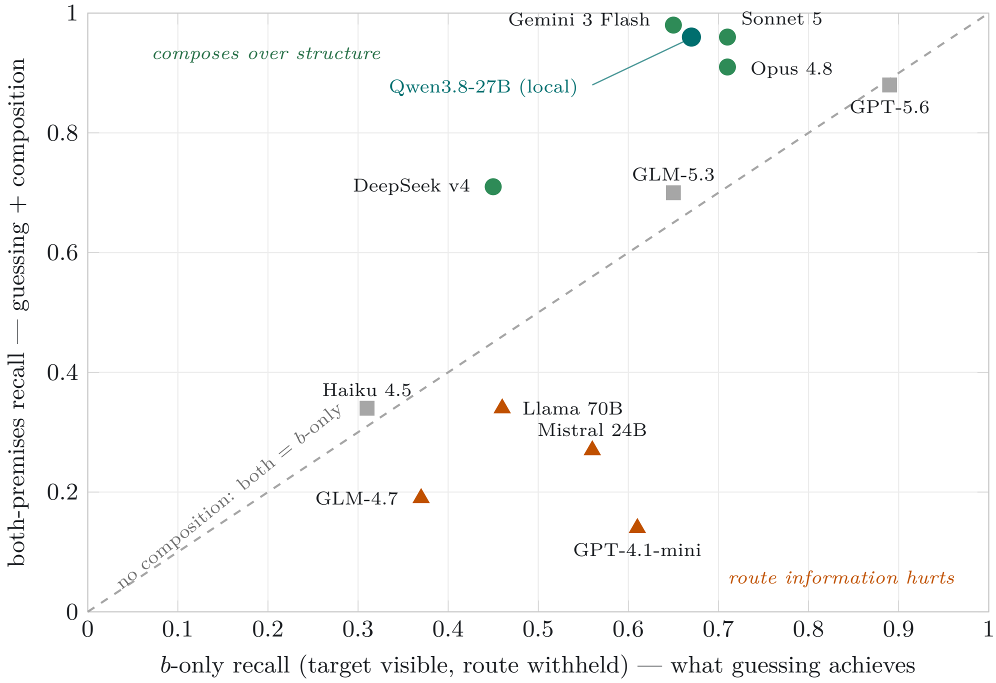

Figure 8: Two-edge composition on one panel. Each mark is a model given the same 150 questions; x is recall when only the answer-bearing premise is served (b-only: the target is verbatim in context but the route from the question's subject is withheld), y is recall with both premises. The dashed diagonal is the no-composition reference; vertical distance above it is the composition-specific component Δ. Green circles: Δ's seeded 10,000-resample bootstrap 95% CI is positive; grey squares: CI spans zero; burnt triangles: CI is negative — for these models the compositional context actively hurts. The three control arms are omitted from the panel because they are uniformly at floor: a-only, matched-irrelevant and raw recall are zero (or $\leq 2/150$) for every model.

- **The delivery studies cannot license reasoning claims.** Every in-domain target is verbatim in the scaffold, and beyond-exposure recovery is negligible ($n_{01} = 3$ of 11,360). The multi-hop design separating composition from exposure and guessing is reported in §10; its limits (two relation types, one corpus, a lexical scorer, single run per arm) remain those of a single-configuration case study.

13 Conclusion

Grounding evaluations where the gold standard derives from the injected corpus should report a copy ceiling: the per-item, model-free recall achieved by a verbatim copy of the shown context. Signed gain over copy re-centres recall against exposure, while the exposure/recovery decomposition distinguishes dropped exposed facts from beyond-exposure recovery, which a bare-model baseline cannot do. Across ten models, gain is uniformly negative. Of 11,360 gold items, only 3 unexposed items are recovered; context utilisation is 0.900–0.955, so beyond-exposure recovery is negligible. A production paired study lifts judged quality by +0.27 pooled and +0.79 where curation is deep. A four-arm control study is consistent with content-specific lift: the placebo's point estimate collapses once verified seed-disjoint. The reusable method is *the ceiling plus a verified-irrelevant placebo*, computed before architecting to distinguish a serving regime, where verbatim retrieval wins, from a synthesis regime, where heavier machinery must be measured above the ceiling rather than the bare model. Value therefore concentrates upstream at curation, where the instrument also serves as a promotion gate for extraction candidates. For a private corpus, the goal was never a model that reasons its way to answers it never saw. It is a model that delivers vetted knowledge faithfully, attributably and cheaply, and an instrument honest enough to prove that is what it does.

For readers outside this literature, organisations increasingly want AI assistants that answer from their own policies, research and technical documentation without sending that knowledge to a cloud provider. This need not be a compromise: a modest local model connected to a carefully curated knowledge graph answered domain questions about as well as the most capable commercial systems because nearly all accuracy came from knowledge placed before it, rather than facts it inferred independently. We establish this using a control often omitted from evaluations: the score achieved by verbatim copying, above which any claimed intelligence must appear. Ten commercial models sit at or below that line for delivery. A separate composition test, where copying cannot help, identifies models that can chain two facts together: several of the largest cannot, while the local model can. A blind quality judgement also caught a pipeline step that was quietly worsening pages before deployment at scale. The practical lesson is to invest in knowledge and curation, serve it faithfully, keep it local, and measure

Table 9: The multivariate bar and this work's status. A deployable private-knowledge system must clear all four *together*; the axes are measured on different instruments, so no single number establishes the conjunction.

Axis	Instrument & evidence	Status
1. In-domain delivery fidelity	Copy ceiling / gain over copy: ten models −0.067 to −0.022; live loom−raw +0.27 pooled, +0.79 arcane	**Measured** (both studies)
2. Out-of-domain non-regression	Judged Δ vs frontier gold: five-model arm all CIs include zero (worst −0.23 [−0.63, +0.05]); live general +0.05 [0.00, +0.13] inside ±0.25 TOST margin	**Measured** (both studies)
3. Beats web search in-domain	Same model grounded in live web results vs curated scaffold, identical question set	Specified; not measured
4. Attribution precision	Per-answer generation + confidence already emitted; precision metric to close "faithful" → "attributable"	Specified; not measured

machine reasoning against verbatim copying.

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